

PySensemakr: Sensitivity Analysis Tools for OLS in Python

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Summary

PySensemakr is a Python package that implements a suite of sensitivity analysis tools for linear regression models based on the omitted variable bias (OVB) framework developed in Cinelli and Hazlett (2020). In observational studies, researchers typically adjust for observed covariates in a regression model and assume that no unobserved confounders remain. Since this assumption is untestable, sensitivity analysis asks how strong unobserved confounders would need to be to alter the conclusions of a study. **PySensemakr** provides tools that make this question precise and easy to answer.

Specifically, the package allows researchers to (i) compute sensitivity statistics for routine reporting, including the partial R^2 of the treatment with the outcome and the robustness value (RV_q), which measures the minimum confounding strength—in terms of the proportion of residual variance explained—required to change a study’s conclusions; (ii) derive bounds on the strength of unobserved confounders using observed covariates as benchmarks, asking, for example, how strong a confounder would need to be relative to an observed covariate such as gender or income to overturn a result; and (iii) produce contour plots and extreme scenario plots that provide a visual summary of the sensitivity of causal estimates to unobserved confounding. **PySensemakr** is built on top of **statsmodels** (Seabold and Perktold 2010) and requires only standard regression output, without assumptions on the functional form of the treatment assignment mechanism or on the distribution of the unobserved confounders.

Statement of Need

The most widely used strategy for drawing causal inferences from observational data is to adjust for observed covariates in a regression model. The key untestable assumption underlying this approach is that there are no unobserved confounders—variables that jointly affect both the treatment and the outcome and that are not included in the model. Since this assumption cannot be verified from the data at hand, the credibility of causal claims rests on how robust they are to potential violations. Sensitivity analysis formalizes this question: rather than assuming confounders are absent, it quantifies how strong they would need to be to change the conclusions of a study.

Cinelli and Hazlett (2020) proposes a modern suite of sensitivity analysis tools based on a partial R^2 parameterization of omitted variable bias. The key idea is to describe the strength of unobserved confounders in terms of the partial R^2 —the share of residual variance they explain—of the confounder with the treatment and the outcome, after accounting for observed covariates. This parameterization has several advantages: (i) it does not require assumptions on the functional form of the treatment assignment mechanism nor on the distribution of the unobserved confounders, (ii) it naturally handles multiple confounders, possibly acting non-linearly, (iii) it exploits expert knowledge to bound sensitivity parameters using observed covariates as benchmarks, and (iv) it can be easily computed using only standard regression output. The methodology lends itself naturally to *routine reporting*—the practice of including sensitivity statistics alongside standard regression tables, much like one reports p -values or confidence intervals—because it produces a small number of interpretable summary measures that can be reported in any empirical paper.

While **sensemakr** implementations already exist for R (Cinelli, Ferwerda, and Hazlett 2024) and Stata,

Python has lacked an equivalent tool, despite being one of the most widely used languages for data analysis across the social sciences, biomedical research, and industry. **PySensemakr** fills this gap, bringing the full **sensemakr** methodology to the Python ecosystem. Its target audience includes applied researchers in the social sciences, epidemiology, and any field where OLS regression is used to estimate causal effects from observational data, as well as data scientists in industry who need to assess the robustness of their findings.

State of the Field

The R version of **sensemakr** (Cinelli, Ferwerda, and Hazlett 2024) has been widely adopted across disciplines—including political science, economics, epidemiology, and education—and has been cited in thousands of empirical studies. A Stata version is also available. **PySensemakr** is a dedicated, full-featured Python implementation of the complete **sensemakr** methodology for OLS regression. It provides the full suite of sensitivity statistics (partial R^2 , robustness values RV_q and $RV_{q,\alpha}$), the complete benchmarking apparatus, all bias-adjustment functions, and all visualization tools (contour plots and extreme scenario plots), with an API that closely mirrors the R version.

Other Python packages provide related but distinct functionality. For instance, general-purpose causal inference libraries may include basic sensitivity checks, but none implements the complete partial R^2 -based framework of Cinelli and Hazlett (2020), which includes sensitivity statistics, formal benchmarking bounds, bias-adjusted inference, and specialized visualizations as a unified toolkit. We chose to build a standalone package rather than contribute to an existing library because the methodology requires a tightly integrated set of functions—from computing sensitivity statistics, to bounding the bias using benchmark covariates, to producing specialized diagnostic plots—that would not fit naturally as an extension of a general-purpose regression or plotting library. The standalone design also allows **PySensemakr** to maintain API parity with the R version, easing adoption for researchers who work across both languages.

Software Design

PySensemakr takes a fitted **statsmodels** **OLSResults** object as input and computes all relevant sensitivity statistics from the standard regression output. A key design decision was to build on **statsmodels** rather than requiring users to re-specify their models, allowing **PySensemakr** to integrate seamlessly into existing regression workflows. The package is organized into the following modules:

- **main:** The **Sensemakr** class serves as the primary interface. It takes a fitted model, a treatment variable, and optional benchmark covariates, and computes sensitivity statistics and bounds. It provides **summary()** and **plot()** methods for reporting results.
- **sensitivity_statistics:** Functions for computing the partial R^2 of the treatment with the outcome, the robustness value RV_q (the minimum confounding strength required to reduce the estimate by a fraction q), and the robustness value $RV_{q,\alpha}$ (which additionally accounts for statistical significance).
- **sensitivity_bounds:** Functions for computing bounds on the bias-adjusted estimates, using observed covariates as benchmarks for the plausible strength of unobserved confounders.
- **sensitivity_plots:** Functions for producing contour plots of bias-adjusted estimates and t -values as functions of the sensitivity parameters $R^2_{D \sim Z | \mathbf{X}}$ and $R^2_{Y \sim Z | \mathbf{X}, D}$, as well as extreme scenario plots.
- **bias_functions:** Core functions for computing the bias, bias-adjusted estimates, standard errors, and t -values.

This modular design mirrors the R version’s architecture, making it straightforward for users familiar with the R package to transition to Python. Internally, all computations rely on closed-form expressions derived from the OVB framework, avoiding simulation or resampling, which makes the package fast and deterministic. All numerical results have been validated against the R implementation to ensure correctness.

Example: Sensitivity Analysis for the Darfur Data

We illustrate a typical workflow using the dataset of Hazlett (2020), who studies the effect of being directly exposed to violence (`directlyharmed`) on attitudes toward peace (`peacefactor`) among Darfurian refugees, controlling for covariates such as age, gender, occupation, past political participation, and household size. The analysis proceeds in three steps: fit an OLS regression, construct a `Sensemakr` object specifying the treatment and benchmark covariates, and then call `summary()` and `plot()` to obtain all results.

```
import sensemakr as smkr
import statsmodels.formula.api as smf

# Load data and fit regression model
darfur = smkr.load_darfur()
model = smf.ols(formula='peacefactor ~ directlyharmed + age + farmer_dar + '
                'herder_dar + pastvoted + hhsize_darfur + female + village',
                data=darfur).fit()

# Run sensitivity analysis
sense = smkr.Sensemakr(model=model,
                        treatment="directlyharmed",
                        benchmark_covariates=["female"],
                        kd=[1, 2, 3])

# Print and plot results
sense.summary()
sense.plot()
```

The unadjusted OLS estimate of the treatment effect is 0.097 ($SE = 0.023$, $t = 4.18$). The `summary()` method reports the key sensitivity statistics. The partial R^2 of the treatment with the outcome is 2.2%, meaning that an extreme confounder explaining 100% of the residual outcome variance would need to explain at least 2.2% of the residual treatment variance to fully account for the observed effect. The robustness value is $RV_q = 13.9\%$: unobserved confounders that explain less than 13.9% of the residual variance of both the treatment and the outcome are not strong enough to reduce the point estimate to zero. The robustness value accounting for statistical significance is $RV_{q,\alpha} = 7.6\%$: confounders explaining less than 7.6% of the residual variance of both the treatment and the outcome cannot bring the estimate to a range where it is no longer statistically distinguishable from zero at the 5% level.

The `summary()` method also reports bounds on the bias-adjusted estimates using observed covariates as benchmarks for confounding strength. Using `female` as a benchmark, if an unobserved confounder were as strongly associated with the treatment and outcome as `female` is ($1 \times \text{female}$), the bias-adjusted estimate would be 0.075 ($t = 3.44$, 95% CI: [0.032, 0.118]). Even at $2 \times \text{female}$, the adjusted estimate remains positive and significant (0.053, $t = 2.60$). At $3 \times \text{female}$, the adjusted confidence interval begins to include zero (0.030, 95% CI: [-0.006, 0.067]), indicating the level of confounding needed to overturn the finding. These benchmarking results allow researchers to make calibrated judgments: rather than asking whether confounding exists in the abstract, they can assess whether confounders plausibly as strong as key observed covariates are likely to be present in their application.

?? shows the contour plots produced by `sense.plot()`. The left panel displays contour lines of the bias-adjusted estimate as a function of the sensitivity parameters $R^2_{D \sim Z | \mathbf{X}}$ and $R^2_{Y \sim Z | \mathbf{X}, D}$; the right panel shows the corresponding t -value contours. The red diamonds mark the benchmark bounds based on `female`. Figure 1 shows the extreme scenario plot, which displays the bias-adjusted estimate under the worst-case confounding scenario as a function of the confounder's association with the treatment, for different assumptions about its association with the outcome ($R^2_{Y \sim Z | \mathbf{X}, D}$ at 100%, 75%, and 50%). Together, these plots allow researchers to communicate the sensitivity of their findings in a single, interpretable figure.

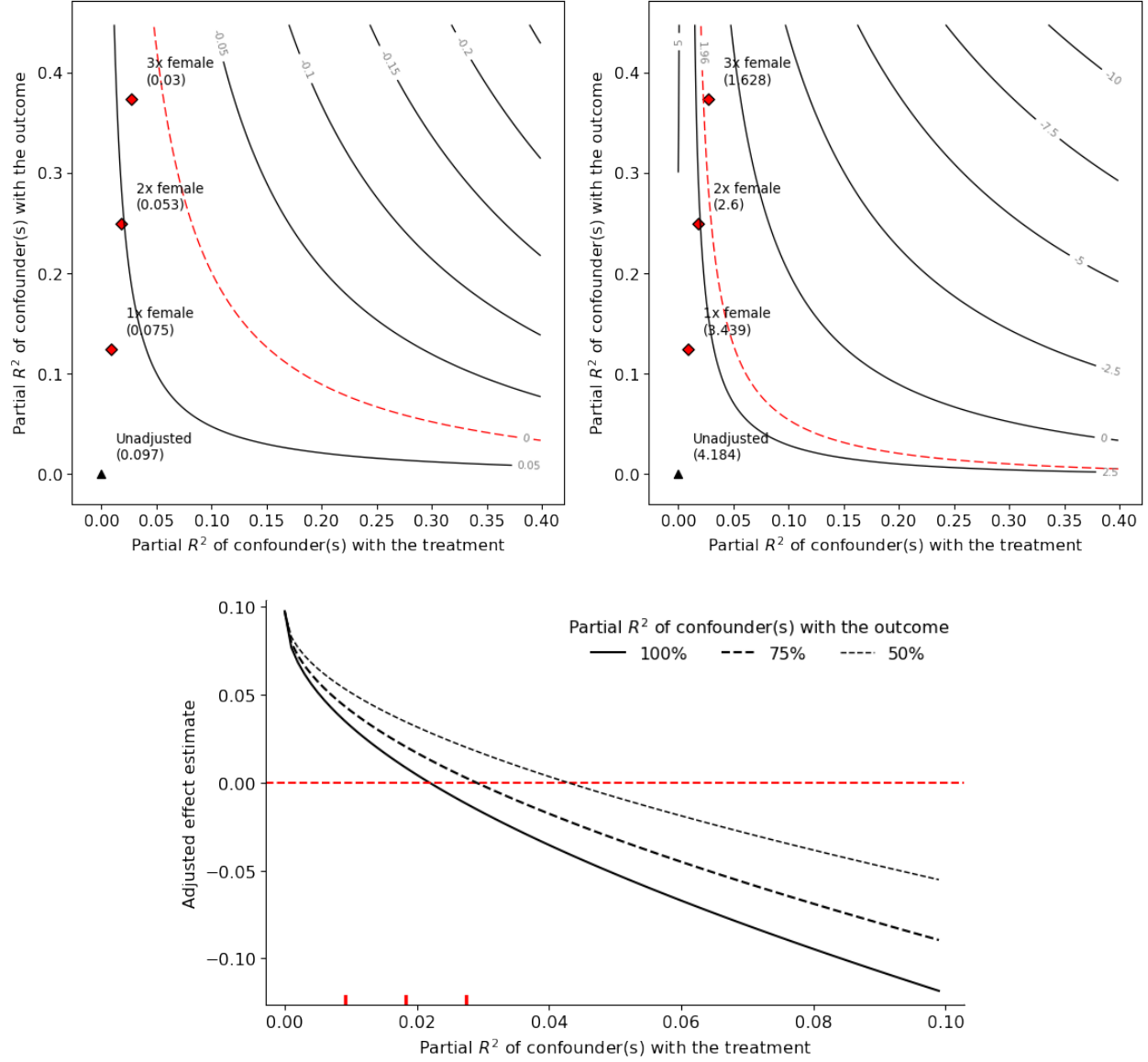


Figure 1: Extreme scenario plot for the Darfur data, showing the bias-adjusted estimate under the worst-case scenario for confounders of varying strength.

Research Impact

The sensitivity analysis methodology implemented in **PySensemakr** (Cinelli and Hazlett 2020) has had broad impact across the empirical sciences. The original theoretical paper has been cited in thousands of empirical studies, and the R implementation of **sensemakr** (Cinelli, Ferwerda, and Hazlett 2024) has been used in published work spanning political science, economics, epidemiology, sociology, education, and environmental science, among other fields. The methodology has also been incorporated into textbooks and graduate-level courses on causal inference, and the robustness value has become a standard reporting measure in several applied disciplines.

By providing a Python implementation, **PySensemakr** extends the reach of these tools to the large and growing community of researchers and data scientists who use Python as their primary computing environment. The package includes a comprehensive test suite validated against the R implementation, ensuring

numerical equivalence across platforms. `PySensemakr` is installable via `pip`, is hosted on PyPI, and includes documentation and example notebooks to facilitate adoption.

AI Usage Disclosure

Generative AI tools were used to assist with code maintenance tasks (e.g., updating deprecated API calls for compatibility with newer versions of dependencies). All AI-generated changes were reviewed and tested by the authors.

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References

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